Experiment 14: Alpha-Beta Pruning

Aim

To write a Python program to implement the Alpha-Beta Pruning algorithm.

Procedure

1. Define terminal node values.
2. Initialize alpha and beta values.
3. Traverse the game tree.
4. Prune unnecessary branches.
5. Display the optimal value.

Source Code

main.py Visualize Run ▶ Output

```
import math

scores = [3, 5, 6, 9]

def alphabeta(depth, node, maximizing, alpha, beta):
    if depth == 2:
        return scores[node]

    if maximizing:
        value = -math.inf
        for i in range(2):
            value = max(value,
                        alphabeta(depth+1, node*2+i, False,
alpha, beta))
            alpha = max(alpha, value)
            if alpha >= beta:
                break
        return value

    else:
        value = math.inf
        for i in range(2):
            value = min(value,
                        alphabeta(depth+1, node*2+i, True,
alpha, beta))
            beta = min(beta, value)
            if alpha >= beta:
                break
        return value

print("Optimal Value:",
      alphabeta(0, 0, True, -math.inf, math.inf))
```

```
Optimal Value: 6
```

Result

Thus, the Alpha-Beta Pruning algorithm was implemented successfully.